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Accurate predictions of mechanical properties using active learning on local chemical configurations: A case study of WC–Co composites

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ABSTRACT

Modern technologies require new materials with a desired set of properties. Key areas of industry use hard and superhard alloys and ceramics. The search for new compositions requires the application of digital materials science techniques. However, computational investigations and screening of these materials require sufficient computational resources, as most industrial materials are polycrystalline, composite, or otherwise structurally heterogeneous, which poses a challenge for computational materials science. We propose a computational method for calculating the elastic and mechanical properties of complex systems with first-principles accuracy by developing and applying a highly efficient procedure using actively learned machine learning interatomic potentials on local chemical configurations (ALLCC). This method is applied to the study of WC–Co composites, where moment tensor interatomic potentials are actively trained on local fragments calculated by density functional theory. As a result, a detailed investigation of changes in the brittle behavior of WC–Co composites as a function of Co content is presented, and complex changes in elastic properties compared to those of WC polycrystals are observed. Simulated stress–strain relationships allow us to determine the main features of the plastic behavior of WC–Co composites under critical tensile deformations. This work establishes active learning techniques, namely, active learning on local chemical configurations, as a powerful and efficient tool for bridging the gap between first-principles accuracy and system-size requirements required to predict and rationally design the mechanical behavior of technologically essential non-crystalline complex solids.

1. Introduction

Most engineering materials used in industrial applications such as mining, machining, cutting, drilling, and manufacturing are polycrystalline, composite, or otherwise structurally heterogeneous rather than perfect single crystals. The most widely used composite is a hard alloy of tungsten carbide and cobalt, usually containing about 94% WC and 6% Co (WC94-Co). The WC–Co composites can be considered one of the hardest materials, in which hard WC grains are surrounded by a Co binder. WC–Co composites have been widely used due to their high hardness and excellent wear resistance [1–3]. Generally, the mechanical properties of WC–Co depend on the size of the WC grains [4–7]. Especially, the specific volume of inter-granular boundaries increases with decreasing grain size, which has a significant effect on the mechanical properties of such composites. Moreover, the experimental synthesis of nanopolycrystalline WC–Co composites is challenging due to the high cost of experiments and the difficulty in establishing a desirable consolidation process to prepare the bulk material from the nanoscale composite powder [8]. Also, the mechanical properties of the WC–Co composite highly depend on the method of synthesis.

An important issue regarding the investigation of structure–property relationships arises. A possible solution is to apply computational techniques for atomistic simulations in order to deepen our knowledge about the relationship between structure and properties, with the aim of achieving desired mechanical properties.

Historically, two families of methods have been employed to describe such systems: empirical interatomic potentials and ab initio calculations. Empirical models, for example, EAM [9], MEAM [10], and ReaxFF [11] – rely on a predetermined analytical expression governed by only a handful of adjustable parameters. These parameters are typically fitted so as to reproduce the basic properties of single crystals as well as the available experimental measurements. For the WC–Co system, the ReaxFF force field [12] is the empirical potential most frequently adopted. A notable limitation, however, is that empirical potentials often fail to capture the intricate interactions arising at inter-granular boundaries, since the local atomic arrangement in these regions can deviate markedly from the ordered crystalline lattice against which the potentials were parameterized.

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Density functional theory (DFT) [13,14] sits at the other end of the spectrum. It is the quantum mechanical method most often relied upon when material properties have to be predicted from scratch. The energies and forces it returns are very accurate, yet the price is steep: in practice, DFT can treat no more than a few hundred atoms at a time, so a direct description of the grain boundary regions stays out of reach. Machine learning offers a way around this bottleneck, and over the last few years, it has quickly become a standard tool for representing interatomic interactions. The aim of such models is to keep the speed of empirical potentials while reaching towards the accuracy of quantum mechanical calculations. Because their functional form is so flexible, machine learning interatomic potentials (MLIPs) can, in principle, match any target potential energy surface, provided that enough parameters are fitted. What really sets them apart from ordinary empirical potentials is the way they encode the local atomic surroundings: in place of a fixed analytical formula, they rely on rich numerical descriptors, as is done in schemes such as GAP [15], MTP [16], NNP [17,18], DeePMD-kit [19], NEP [20], and others. Over the past decade, the application of machine-learning techniques to atomistic simulation has grown markedly [21–27]. The parameters of an interatomic potential are typically determined during the training stage by minimizing the discrepancy between the predicted energies and forces and the corresponding values obtained from first-principles calculations for configurations included in the training dataset.

Nevertheless, whenever a configuration sampled during a simulation departs appreciably from those contained in the training set, the potential is forced to extrapolate, and this can introduce substantial prediction errors. Overcoming this drawback requires MLIPs that are able to recognize in advance the configurations for which such extrapolation is expected to take place. One effective strategy is the *learning-on-the-fly* approach [28], which dynamically augments the training dataset and helps maintain reliable energy and force predictions throughout the simulation. A related idea was put forward by Zhang and co-workers [29] in their Deep Potential GENerator (DP-GEN). There, the training set is grown step by step, on the fly, with very little input from the user. The goal is to reach deep learning models of the potential energy surface that stay accurate everywhere and to get there without the heavy cost that data generation and model fitting usually bring. Building on this line of work, here we develop an active-learning methodology for MLIPs based on moment tensor potentials (MTPs) [16], which automatically identify and extract local extrapolative configurations using the D-optimality criterion as implemented in the *MLIP* package [30]. These local structures are subsequently transformed into compact periodic configurations with physically consistent periodic boundaries, making them suitable for DFT calculations. We then apply this active learning on the local chemical configurations (ALLCC) scheme to follow how the mechanical properties of WC–Co shift with grain size and with cobalt content. The numbers it returns stay very close to those from direct DFT. Beyond that, we check how reliably the procedure spots the locally extrapolative environments during a run and turns them into periodic cells that DFT is able to handle.

2. Methods

2.1. Generation of WC–Co structure

The WC–Co composite structures were built using the Voronoi tessellation approach [31–33] available in AtomsK [34]. In this scheme, a periodic simulation domain is partitioned into a prescribed number of grains possessing randomized shapes and crystallographic orientations. First, the WC polycrystalline structures with a different number of grains (2 and 3) are constructed. After the WC region of the grain boundary is replaced with Co atoms. These Co atoms serve as a binder between WC grains. Overall, the computational domain has a cubic shape, which size is $10 \times 10 \times 10 \text{ nm}^3$ and contains up to 50 000

atoms. In such constructed composites, the thickness of the Co binder varies from 2 to 5 atomic layers. The following notation of WC(100-X)-Co(X) is used to identify the concentration of Co in the composite. So, a pure WC polycrystal is denoted as WC100. Thus, in total, we consider 8 different compositions of the composites, namely WC100 (or just WC), WC90-Co10, WC87-Co13, WC84-Co16, WC81-Co19, WC78-Co22, WC75-Co25, and WC72-Co28.

To perform the simulations of tensile deformation and to obtain stress–strain relations for WC–Co composites with different Co content, the cylindrical WC–Co rods are constructed. Each rod has a length of 200 Å and a base diameter of 50 Å, and consists of 30 000 atoms. Such cylinders are placed in a simulation box measuring $100 \times 100 \times 200$ Å. Periodic boundary conditions are applied only along the tensile axis Z , while the other directions are considered to be free.

2.2. DFT dataset generation

The initial dataset contains the results of *ab initio* molecular dynamics of WC (P6₃m2, 187 space group), Co (P6₃/mmc, 194 space group), and a small WC–Co heterostructure (see supporting information). Density functional theory (DFT) was used to perform AIMD calculations.

For the DFT part, we used the projector augmented wave method (PAW-DFT) [14,35] as built into VASP [36–39]. Exchange and correlation were described at the GGA level with the Perdew, Burke, and Ernzerhof functional (GGA-PBE). Every element was given its own PAW potential, chosen according to how many valence electrons it contributes, namely 6 for tungsten, 4 for carbon, and 9 for cobalt. A plane wave cutoff of 500 eV, together with Methfessel and Paxton [40] smearing of the occupation numbers, proved sufficient to converge the total energy. The Brillouin zone was sampled with a Γ -centered $8 \times 8 \times 8$ mesh of k points. Ionic positions were optimized using the conjugate gradient algorithm provided by the code, and the relaxation was stopped once the largest residual force had fallen below $0.01 \text{ eV}\text{\AA}^{-1}$.

2.3. Active learning on local chemical configurations

Transferability constitutes one of the principal difficulties in the use of MLIPs. Because these models obtain energies and forces by interpolating over the training data, the dataset must adequately span the range of configurations that the simulation will actually visit. Otherwise, the model is forced to extrapolate, which typically results in a substantial loss of prediction accuracy. Consequently, the practical application of MLIPs requires a reliable method for detecting extrapolative configurations, commonly referred to as active learning techniques.

During molecular mechanics simulations, the extrapolation grade is evaluated for the local atomic environment of every atom using the D-optimality criterion implemented in the MLIP package. Whenever the extrapolation grade exceeds the prescribed threshold, the corresponding local atomic environment is identified as extrapolative and selected for further processing. Once an extrapolative configuration is identified, it can be computed using DFT and subsequently incorporated into the training set. This procedure expands the domain covered by the MLIP and reduces the likelihood of future extrapolation.

Different MLIP frameworks employ different approaches for extrapolation detection. Gaussian process-based MLIPs estimate the prediction uncertainty (variance) and use it as an indicator of extrapolation [41]. Neural network-based MLIPs rely on the disagreement between members of a model committee to identify inputs located outside the training domain [42]. In the case of MTPs, the extrapolation grade is quantified using the maximal volume principle. Specifically, the extent to which the training set spans the configuration space is evaluated using the MaxVol algorithm [43]. The extrapolation metric can be computed both for an entire configuration and for the local

atomic neighborhoods of individual atoms. This enables the identification of local regions where the accuracy of force predictions may be insufficient. Such a capability is particularly valuable for simulations involving large atomic systems.

The proposed ALLCC methodology does not explicitly classify local atomic environments according to their structural origin, such as WC grains, Co binder, WC/WC grain boundaries, or WC/Co interfaces. Instead, every local atomic environment is assigned an extrapolation grade. Configurations with the highest extrapolation grades are automatically selected regardless of their structural origin. Consequently, grain boundaries, phase boundaries, defect-rich regions, and crystalline grains are sampled naturally whenever they contain previously unseen local environments. This eliminates the need for manual identification of different interface types while ensuring that all relevant atomic environments are represented in the training dataset.

However, obtaining *ab initio* reference data for large configurations presents a significant challenge, as DFT calculations for such systems are often computationally infeasible. To address this issue, extrapolative local configurations must be extracted from the large system in a form suitable for DFT calculations, which usually contains no more than a few hundred atoms.

Earlier studies [44,45] isolated such extrapolative atomic environments and handled them as non-periodic atomic clusters prior to the DFT calculations. While this approach is appropriate for systems containing free surfaces, the present work focuses exclusively on bulk configurations. In such cases, including fragments with artificial free surfaces in the training set would undesirably extend the training domain into physically irrelevant regions and could ultimately reduce the accuracy of the potential. Therefore, in this study, we extend a previously proposed approach [46], which has already been successfully applied for investigations of diamond polycrystals. After identifying and extracting a local extrapolative configuration, a cubic fragment centered on the extrapolative atom is extracted from the large-scale structure. The cube size is selected such that its half-length is equal to the MTP cutoff radius, ensuring that the complete local environment contributing to the interatomic interactions is preserved. DFT calculations of energies, forces, and stresses are then performed for the resulting periodic configuration.

A schematic illustration of the procedure used for detecting and extracting new extrapolative local environments is presented in the Results section.

2.4. Mechanical properties calculations

All molecular mechanics simulations by using MTPs were carried out within the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [47]. The independent elastic constants of the composites were obtained by means of the conventional atomistic protocol detailed in Ref. [48]. For each WC–Co composite, the atomic coordinates, together with the cell shape and volume, were first relaxed until the pressure reached zero or near-zero values; subsequently, finite strains of either sign and magnitude of about 1% were imposed along all symmetry-inequivalent directions. Further, the stress tensor for a set of small applied strains ϵ_{ij} is computed using a linear relation to extract the elastic constants C_{ijkl} :

$$\sigma_{ij} = \sum_{kl} C_{ijkl} \epsilon_{kl} \quad (1)$$

where σ_{ij} are components of the stress tensor. After imposing the strain, the atomic positions are relaxed while keeping the supercell shape fixed to obtain the equilibrium configuration. The resulting stress tensors for each relaxed, strained configuration are then evaluated. Overall, components of the elastic tensor C_{ij} were obtained by applying finite-difference formulas to the stress–strain data.

Elastic tensors were determined for a range of WC–Co structures spanning different Co fractions. The calculations of the bulk modulus B ,

shear modulus G , Young's modulus Y , Poisson's ratio ν , and Pugh ratio B/G are performed by using elastic constants. The Vickers' hardness calculation is performed by using Chen's model [49]. Statistical noise was held in check with k-means clustering at $k = 8$. We kept drawing new samples until the spread of the k-means estimates fell under 5% of their average.

To calculate the tensile properties of WC–Co, zero-temperature simulations are performed. First, the energy minimization of the systems at zero temperature and zero pressure is provided. After equilibration, the tensile strain is applied along the periodic z -direction with a specific strain rate of 10^{-6} Å per step. The box is changed by 30 Å along the periodic z -direction, before which the WC–Co had reached the yield strength. We performed the tensile deformation on the WC–Co composite system rods with different contents of Co binder, namely WC100 (or just WC), WC90–Co10, WC87–Co13, WC84–Co16, WC81–Co19, WC78–Co22, WC75–Co25, and WC72–Co28. We obtained the stress–strain curve and yield strength of WC–Co from the tensile deformation simulations.

3. Results

3.1. Active learning procedure for MTP for WC–Co simulations

The initial training configurations for MTPs are obtained from high-temperature (over 1000 K) *ab initio* molecular dynamics simulations (AIMD), causing atoms to undergo intense thermal vibrations. AIMD simulations for single-crystal structures of WC and Co and for WC–Co heterostructures containing up to 100 atoms are performed. The total simulation time of each AIMD is 2 ps. This data is used to train the initial MTP(1) (see step (1) in Fig. 1) in a passive manner with atomistic configurations sampled from AIMD, as on-the-fly active learning of the MTP from scratch is not computationally efficient.

After passive learning, MTP(1) then served to evaluate the elastic tensors of WC and Co single crystals and of WC–Co heterostructures within molecular-mechanics (MM) simulations (step (2) in Fig. 1). Throughout these runs, extrapolative configurations were selected on the fly using the one-extrapolation-threshold (γ_{break}) strategy [30] (step (2a) in Fig. 1). Whenever γ_{break} is exceeded, the extrapolation grade is so large that energies, forces, and stresses can no longer be predicted reliably; the elastic-tensor calculation is therefore halted, and MTP(1) is retrained. Guided by our previous experience [46], γ_{break} was fixed at 11, which strikes a favorable compromise between the accuracy of the potential and the frequency of retraining. The active-learning procedure is described in full in Ref. [30]. After reaching the simulation-breaking condition ($\gamma > \gamma_{break}$), we identify all extrapolated configurations and select them to expand the training set. Further, DFT calculations of energy, force, and stress are performed for selected configurations; see step (2b) in Fig. 1. This allows us to obtain new configurations for further updates of the training set (step (2c)). A selection process is necessary to create a new active set from a pool of extrapolated configurations.

In the next step, these new configurations join the original training set, and MTP(1) is retrained on the enlarged set to give MTP(1+), as shown in step (2d) of Fig. 1. This active learning strategy for the ideal small single crystal structures is necessary to increase the stability of MTP. After several active learning loops (generally not more than 50 times), a new MTP(2) is obtained, see step (2e) in Fig. 1. MTP(2) can now describe interactions in the WC, Co, and WC–Co required for simulations of WC–Co composites. Nevertheless, actively trained MTP(2) shows no reasonable performance during the molecular mechanics simulations of big and realistic WC–Co structures containing more than 50 000 atoms. The MTP(2) is tested during MM using a big WC–Co composite structure with active extrapolation selection. During simulation, local chemical configurations with an extrapolation grade higher than termination grade ($\gamma \gg \gamma_{break}$) are detected. The extrapolation grade during simulations becomes too large (exceeds

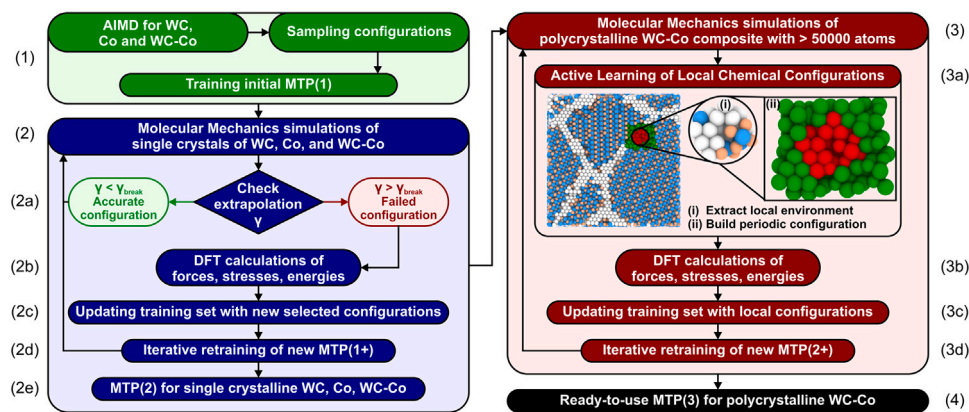


Fig. 1. General procedure of active learning on local chemical configurations using MTP. (1) Passive learning stage that includes the collection of the initial training set, sampling, and training of MTP(1). (2) Active learning (AL) scheme for WC, Co single crystals, and WC-Co small heterostructures involving DFT calculations of small configurations. (3) AL based on local chemical configurations (ALLCC) for the WC-Co composite structures with more than 50 000 atoms. (4) Obtained a highly accurate MTP(3). In the inset, white balls are cobalt, blue — tungsten, orange — carbon. The red color highlights the local configuration extracted from the polycrystal, while the green color is the outer shell added to build a periodic configuration for DFT calculation.

100). Therefore, we have added and performed active learning of MTP(2) on the local chemical configurations, step (3) in Fig. 1. During the MM simulation of the WC-Co composite, we extract a cubic box containing the local chemical (or atomic) configurations from the large extrapolated configuration, see step (3a) in Fig. 1. Such an extrapolative configuration consists of a central atom plus every atom that falls inside its cutoff sphere, here taken with a radius of 5 Å, see (i) in the inset to step (3a). The constructed local chemical configuration extracted from polycrystalline material, typically a cubic cell (side length $l = 11$ Å with 0.5 Å the additional vacuum layer), contains around 100 atoms. Next, this atomic configuration expanded into a periodic structure with the energy, force, and stress calculated by using DFT (steps (3a,b) in Fig. 1). The subsequent stages consist of incorporating the new local configurations into the training set (step (3c)) and retraining the potential to obtain MTP(2+) (step (3d)). Because only the extracted fragments enter the DFT calculations, this scheme removes the necessity of treating the full composite structure at the first-principles level during active training. Accordingly, after the first active-learning iteration is finished, the MM simulation of the elastic tensor is resumed with the updated MTP(2+) and proceeds until the extrapolation grade again reaches its critical value or all configurations have been processed. By this point, upwards of 20 new atomic configurations had been identified, extracted, computed with DFT, and added to the training set. Each pass of this kind broadens the region the potential can describe and makes the MTP more transferable so that extrapolation events grow both rarer and milder as the procedure goes on. Finally, during the calculation of the elastic tensor of a one WC-Co polycrystal, the MTP(3) is obtained using the ALLCC scheme with the accuracy required for further calculation of the mechanical properties of WC-Co composites with DFT-like accuracy, see step (4) in Fig. 1; The potential for the WC-Co composite is ready to use. The scheme, similar to active learning on local chemical configurations, was used previously in Refs. [46,50] to study the mechanical properties of diamond polycrystals and diffusion mechanisms in SiC structures.

The configuration space of the obtained training set is examined using t-distributed stochastic neighbor embedding (t-SNE) visualization. This method allows us to examine how configurations from different learning strategies are distributed relative to each other, see Fig. 2a. This dimensionality-reduction technique enables an intuitive inspection of cluster compactness, separability, and potential overlap that may not be apparent from quantitative metrics alone.

t-SNE analysis was performed on the Smooth Overlap Atomic Positions (SOAP) descriptors computed for individual atomic environments, such that each point in the low-dimensional embedding represents a single atom characterized by its local neighborhood within a $r_{cut} = 5$ Å

cutoff radius, the maximum number of radial basis functions $n_{max} = 12$ Å, and the maximum degree of spherical harmonics $l_{max} = 10$ Å [15,51].

As shown in Fig. 2a by green color, configurations obtained through passive learning form relatively dense and coherent clusters, indicating consistent feature representations. New structural information added by active learning for single crystals (step (2) in Fig. 1) is shown by the blue color in Fig. 2a, forming a clearly separated cluster, suggesting that these configurations occupy a distinct region of the feature space. Active learning produces more fragmented yet well-defined subclusters, reflecting increased diversity in the selected configurations. Notably, ALLCC forms smaller, tightly grouped clusters positioned near related regions (see reg regions in Fig. 2a), indicating that locally focused sampling captures specific substructures while maintaining proximity to the broader data manifold. The t-SNE optimization converged to a final Kullback–Leibler (KL) divergence of 1.52, indicating good preservation of local neighborhood relationships in the low-dimensional embedding. The mean bandwidth parameter ($\langle \sigma \rangle = 0.832$) reflects stable local density estimation across samples, suggesting reliable manifold representation of the training data. Thus, the training set covers a broad range of atomic configurations generated during the three stages of training the MTP, exhibiting distinct characteristics without significant overlap.

3.2. Performances of MTPs for the WC-Co composite

The accuracy of the developed MTPs (MTP(1), MTP(2), and MTP(3)) for the WC-Co composite was evaluated after the third stage (shown in Fig. 1). To quantify the predictive performance of the potential, the correlations between the DFT reference data and MTP predictions were analyzed for total energies, atomic forces, and stress tensors on the whole obtained training set. For each quantity, the maximum absolute error (MaxAE), mean absolute error (MAE), and root-mean-square error (RMSE) were calculated.

The assessment was performed for all three stages of the training workflow: (i) passive learning based on the initial AIMD dataset (MTP(1)), (ii) active learning on WC and Co single crystals together with small WC-Co heterostructures (MTP(2)), and (iii) active learning on local chemical configurations (ALLCC), resulting in the final MTP(3).

The obtained accuracy metrics are summarized in Table 1. A systematic improvement of all error measures is observed at every stage of the training procedure. Passive learning provides only a preliminary description of the potential energy surface and exhibits relatively large deviations from the DFT reference data. The subsequent active learning stage significantly improves the accuracy by extending the training domain with extrapolative configurations encountered in simulations

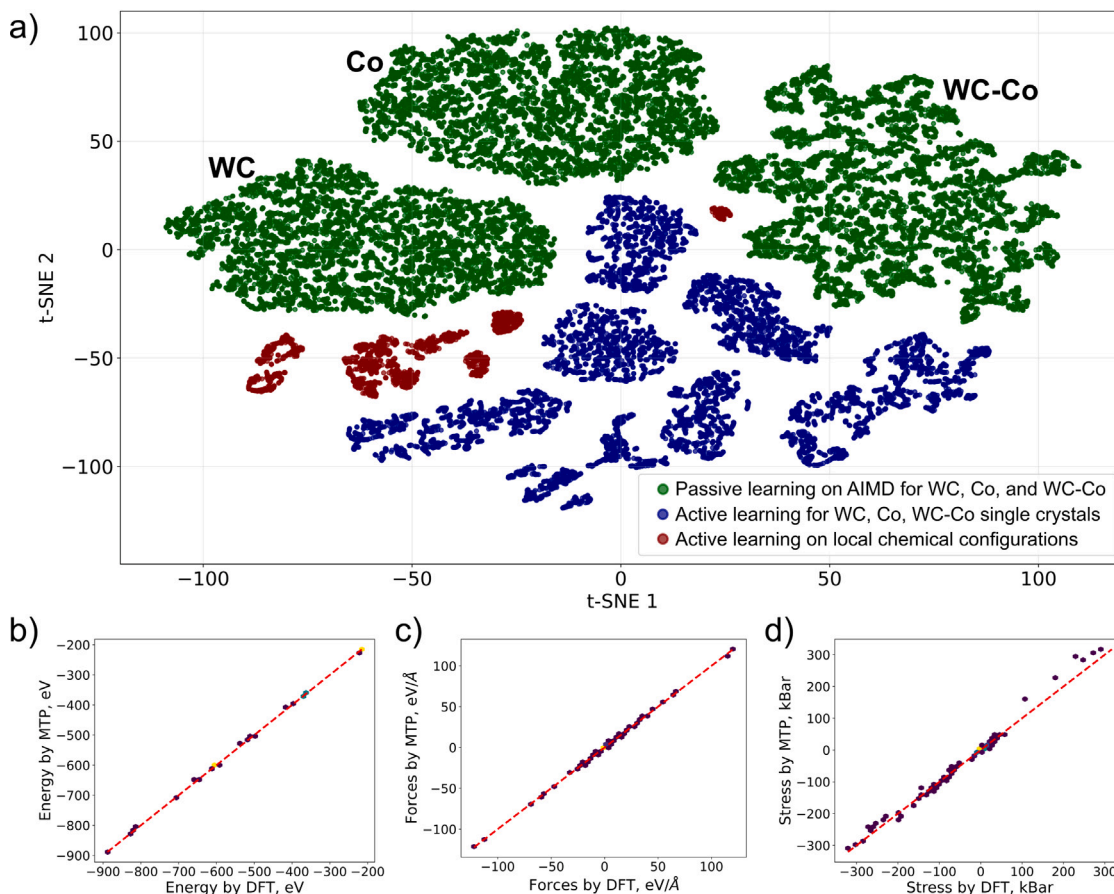


Fig. 2. (a) The t-distributed stochastic neighbor embedding (t-SNE) visualizations of the training set colored by steps of MTP learning, namely, green is for passive training on AIMD data, blue is for active learning on single crystals, and red is for ALLCC data. Calculated by DFT and fitted by MTP values of total energy (b), forces (c), and stresses (d) obtained for the MTP(3) trained on local chemical configurations for the WC–Co composite. The hexbin colors encode local point density, with darker shades indicating higher density and lighter shades indicating lower density.

Table 1

Accuracy metrics for MTPs obtained after different stages of the training procedure.

Quantity	Metric	MTP(1) Passive learning	MTP(2) Active learning	MTP(3) ALLCC
Energy (eV)	MaxAE	1.549×10^3	1.09×10^2	7.0×10^{-2}
Energy (eV)	MAE	2.861×10^2	3.44×10^1	1.5×10^{-3}
Energy (eV)	RMSE	6.24×10^2	5.64×10^1	5.5×10^{-3}
Force (eV Å ⁻¹)	MaxAE	1.61×10^5	1.035×10^4	2.5
Force (eV Å ⁻¹)	MAE	1.84×10^4	1.15×10^3	1.5×10^{-1}
Force (eV Å ⁻¹)	RMSE	5.6×10^4	2.08×10^3	2.4×10^{-1}
Stress (kBar)	MaxAE	4.14×10^6	2.56×10^5	13.4
Stress (kBar)	MAE	1.66×10^5	1.24×10^4	1.4
Stress (kBar)	RMSE	7.61×10^5	7.53×10^4	1.7

of ideal crystalline systems. The largest improvement is achieved after the ALLCC stage, where locally extrapolative atomic environments extracted from large WC–Co composites are incorporated into the training set. As a result, the final MTP(3) reproduces DFT energies, forces, and stresses with high accuracy, demonstrating excellent transferability to the complex atomic environments present in realistic WC–Co composites.

The parity plots comparing DFT and MTP predictions for the final MTP(3) are shown in Fig. 2b–d. The obtained results demonstrate that active learning on local chemical configurations substantially improves the predictive capability of the potential and enables reliable

simulations of large-scale WC–Co composite systems with near-DFT accuracy.

3.3. Mechanical properties of WC–Co composite

The calculations of elastic properties, together with the simulation of tensile deformation by using MTP(3) on local chemical configurations, are performed for WC–Co composites with different Co contents, namely WC100 (or just WC), WC90–Co10, WC87–Co13, WC84–Co16, WC81–Co19, WC78–Co22, WC75–Co25, and WC72–Co28.

The elastic moduli are evaluated from the components of the elastic stiffness tensor using the Voigt–Reuss–Hill averaging scheme. For each composition, between 15 and 30 independent structures are generated in order to quantify the variability of the calculated elastic moduli, elastic-property ratios, and hardness values. The corresponding numerical data are provided in the Supporting Information (Fig. S1).

For every composition, the sample mean M and sample variance σ are determined from the ensemble of generated structures. Additional structures were generated, and their elastic properties were calculated until the statistical uncertainty of the estimated quantities was reduced below 5%.

The content of Co in a WC–Co composite can be varied by changing the number of WC grains and the thickness of the Co binder between WC grains. Elastic constants, bulk (B) and shear moduli (G), Young's modulus (Y), Pugh ratio (B/G), Poisson's ratio ν , and Vickers hardness (H_V) are calculated for the WC–Co composite (periodic structure) containing up to 50 000 atoms.

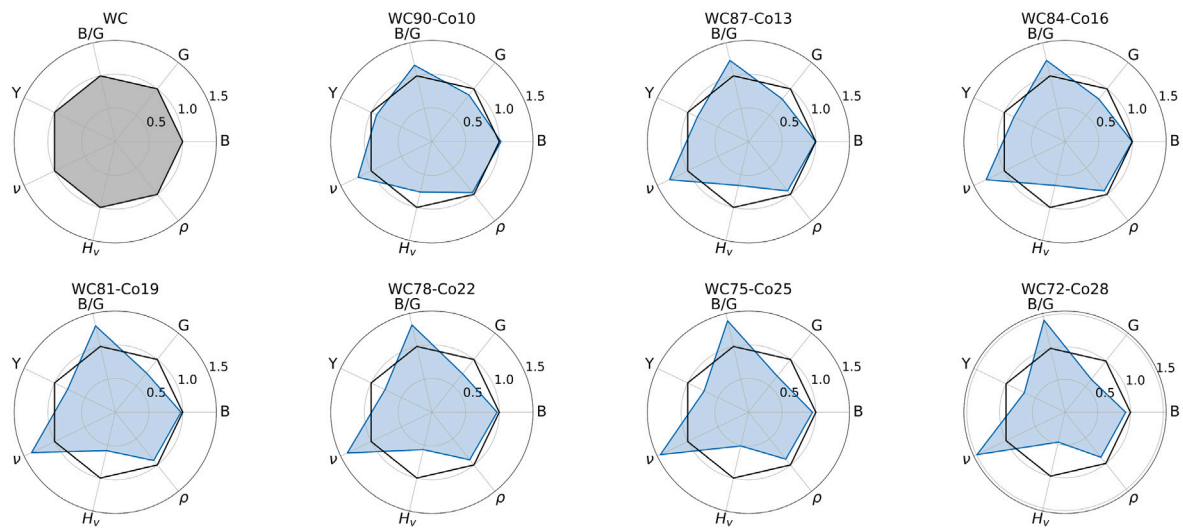


Fig. 3. Star plots of the mechanical properties (bulk (B), shear (G), and Young (Y) moduli, Pugh ratio (B/G), Poisson's ratio, ν , Vickers hardness H_V) and density (ρ) of polycrystalline WC (gray color) and WC-Co composites (blue color) with different Co content.

Obtained numerical results of the calculations of elastic properties are summarized in Table S1 and shown in Fig. 3. Star plots in Fig. 3 provide a visual fingerprint of the mechanical behavior, comparing the polycrystalline WC against the WC-Co composites with varying Co content. The axes represent the bulk modulus (B), shear modulus (G), Young's modulus (Y), Pugh ratio (B/G), Poisson's ratio (ν), and Vickers hardness (H_V). One should note that the shown values of all mechanical characteristics are normalized to polycrystalline WC values (no Co additives). So, each WC-related property has a value of 1, and the values that differ from 1 indicate an increase or decrease relative to polycrystalline WC. Upon the addition of cobalt, a distinct evolution in the mechanical profile is observed. As the Co content increases, there is a noticeable contraction along the axes representing hardness (H_V), Young's modulus (Y), and the shear modulus (G). Conversely, there is a radial expansion along the axes representing the Pugh ratio (B/G) and Poisson's ratio (ν). The bulk modulus of WC-Co drops only a little as more cobalt is added, by roughly 10% and does not exceed the value of the bulk modulus of polycrystalline WC. The decrease in bulk modulus with increasing Co concentration in WC-Co composites is mainly due to the introduction of the softer Co binder between hard WC grains. As the Co content increases, the overall stiffness of the composite decreases because Co has a significantly lower bulk modulus than WC. Consequently, the composite becomes more compressible, leading to a reduction in the effective bulk modulus. A similar tendency of decreasing the values of shear modulus with increasing Co content is also observed. The decrease in G (over 25%) with increasing Co concentration in WC-Co composites occurs because Co is a softer, more ductile binder phase with much lower shear resistance than the hard WC grain. As the Co binder content increases, the composite contains more of this ductile phase, thereby reducing resistance to shear deformation and promoting easier dislocation motion. This behavior is associated with increased plasticity and ductility of the material, leading to a lower G. The decrease in bulk and shear moduli with increasing Co content leads to an increase in both the Pugh ratio (B/G) and Poisson's ratio. A higher Pugh ratio generally indicates enhanced ductility and a greater tendency for plastic deformation in the composite. Similarly, the increase in Poisson's ratio reflects a higher degree of atomic mobility and reduced resistance to shear deformation. As the Co binder increases, the composite becomes mechanically softer due to the lower stiffness of Co compared to WC. This increase in the ductile phase promotes plasticity in the material. Consequently, the Vickers hardness of the composite decreases with increasing Co

concentration (up to 19.6 GPa). A similar dependency of hardness on WC-Co content is reported in Ref. [7].

This progressive change in the star plot morphology reflects the increasing metallic character of the composites. The increase in B/G and ν signifies a transition towards enhanced ductility and plastic deformation capability, as the cobalt binder facilitates slip. However, this gain in toughness comes at the expense of the overall stiffness and hardness, as evidenced by the reduction in the Y, G, and H_V . This will be clearly seen from the simulations of tensile deformations.

Cylindrical models of WC-Co composites are considered to study tensile stress-strain relationships depending on the Co content, as shown in Fig. 4a. Numerical values of the obtained mechanical characteristics are summarized in Table S2 in the Supporting Information. One can note that the ultimate tensile strength (UTS) of a pure WC polycrystal is 21.4 GPa with a critical strain of $\sim 5\%$, while the yield strength is 21.1 GPa. The dependence of UTS on the Co content is presented in the inset of Fig. 4b. The UTS value decreases as the Co content increases from 21.4 to 10.1 GPa (53% decrease), while the critical strain sufficiently increases. The inverse relationship between UTS and critical strain illustrates the classic strength-ductility trade-off where the incorporation of the softer Co phase improves plastic deformability at the expense of ultimate tensile strength. The Young's modulus, derived from the slope of the linear elastic region, shows a monotonic decrease with increasing Co content, reflecting the replacement of the hard WC grain by the more compliant Co binder. The elastic regime becomes much wider, and the transition to yielding starts later. At the same time, the critical strain increases from 5% (polycrystalline WC) to almost 30% (28% Co content in WC-Co composite), see the green curve for the WC72-Co28 structure in Fig. 4b. At larger Co content, the stress-strain behavior begins to resemble that of a plastic material. This transition in mechanical behavior is attributed to the growing contribution of the ductile Co phase, which accommodates plastic deformation through dislocation activity, whereas the WC polycrystals deform predominantly elastically with limited plasticity before failure. Higher Co content leads to an enhancement of the plasticity of WC-Co content, and it behaves more like a ductile material. The progressive evolution from brittle (WC-dominated) to ductile (Co-dominated) fracture behavior underscores the role of the metallic binder in delocalizing deformation and suppressing catastrophic failure. This tendency of becoming more ductile in WC-Co composites is reported in theoretical studies [4,52,53] and in experimental works [54–56].

Moreover, this indicates the ability of our model (and overall, the ALLCC training scheme) to accurately evaluate the evolution of

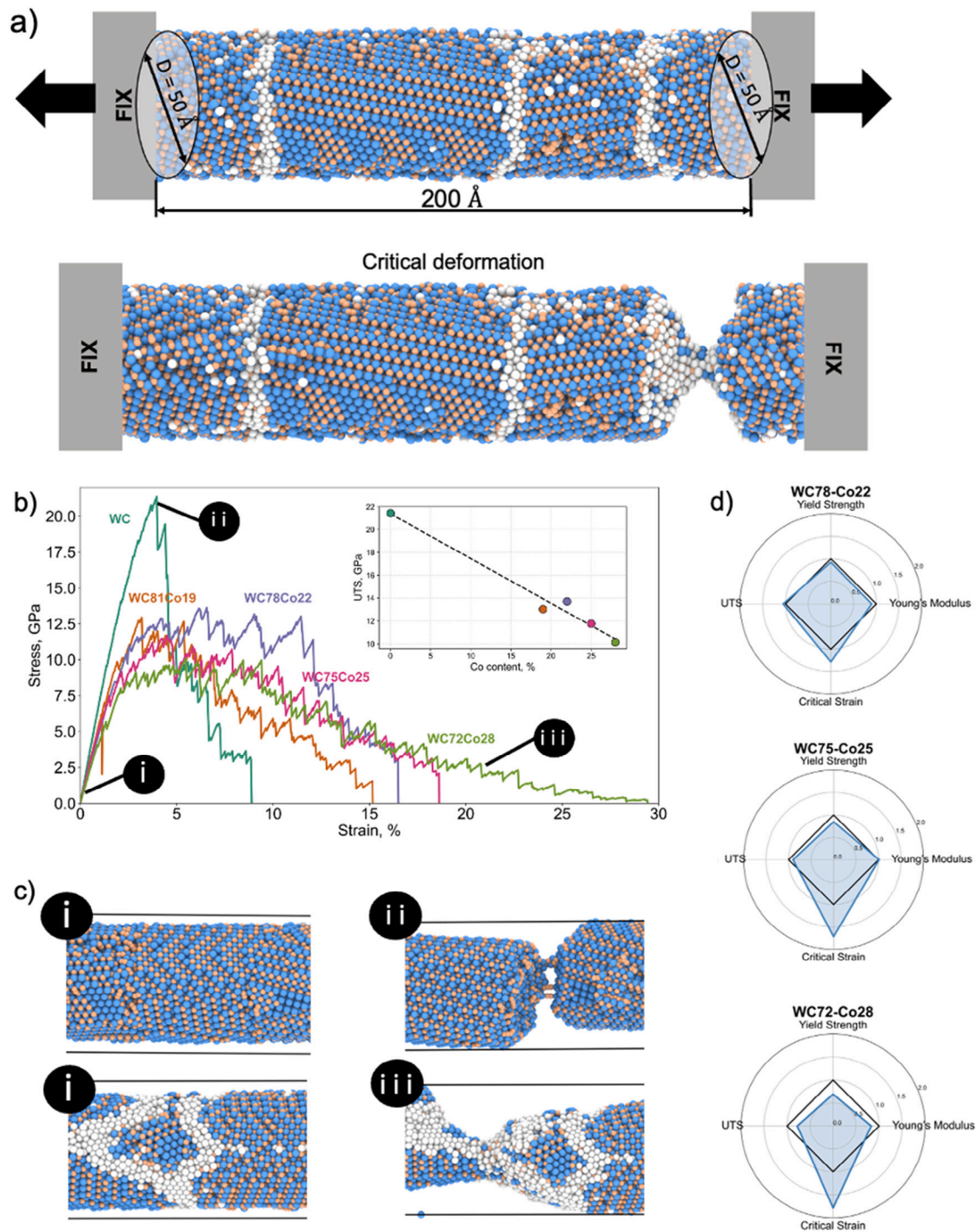


Fig. 4. (a) The principal scheme of simulation including the initial undeformed state and the state under critical deformation of the WC-Co composite. (b) Stress-strain relations calculated for WC polycrystalline and WC-Co composites with different Co contents. Roman numerals on the plot correspond to the deformed structures. (c) Atomic structure of the initial undeformed structure (i) of the WC polycrystals and the WC72-Co28 composite, (ii) is the atomic structure at the critical strain of the WC polycrystals, and (iii) is the atomic structure of the WC72-Co28 at the critical strain. (d) Star plots of the mechanical properties obtained from stress-strain simulation, including UTS, yield strength, Young's modulus, and critical strain of WC-Co composites with respect to WC81-Co19 (lowest Co content considered in the work).

the mechanical properties of complex multi-component non-crystalline materials by machine learning-based simulations. Our approach can then be used for simulations of even more complex materials with binders, which opens the way for accurate prediction of properties of vital instrumental materials for different industrial areas.

4. Conclusion

In this study, we built a practical scheme for training machine learning interatomic potentials on local chemical configurations that are evaluated with first-principles methods, with the aim of probing the mechanical properties of complex, heterogeneous materials. Combinations of machine learning potentials with an active learning strategy focusing on local chemical configurations allow simulations of large composite systems based on information about small periodic configurations suitable for DFT calculations. This enables the training of highly accurate and transferable moment tensor potentials while avoiding the prohibitive computational costs of direct first-principles simulations.

Using this approach, we systematically investigated the mechanical behavior of WC–Co composites with varying Co binder content, revealing clear trends in their elastic properties and brittle-to-ductile behavior. Our results show that increasing Co concentration reduces stiffness, strength, and hardness, while simultaneously enhancing ductility and plastic deformation capability, trends that are quantitatively captured through calculated stress–strain relations and provide detailed atomistic insight into the structure–property relationships governing these composites. The demonstrated methodology not only enables a comprehensive analysis of WC–Co systems but also establishes a generalizable framework for the study of a wide range of instrumental materials, from ceramics and cermets to nanostructured composites, opening new ways for the predictive design of mechanically optimized heterogeneous systems.

CRedit authorship contribution statement

Faridun N. Jalolov: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Viktor S. Baidyshev:** Project administration, Methodology, Investigation, Data curation. **Alexander G. Kvashnin:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.commatsci.2026.114962>.

Data availability

The data that support the findings of this study are available at the following link: <https://gitlab.com/faridunjalolov/mtp-for-wc-co-composite-system.git>.

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